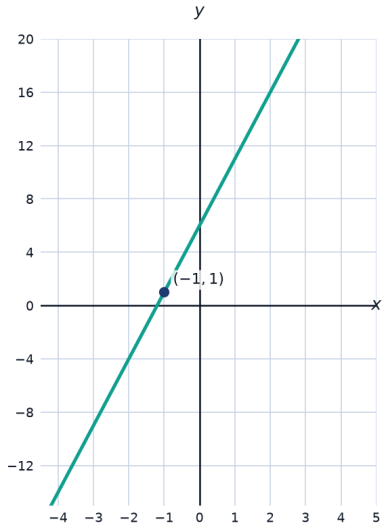
Radicals & Exponents

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

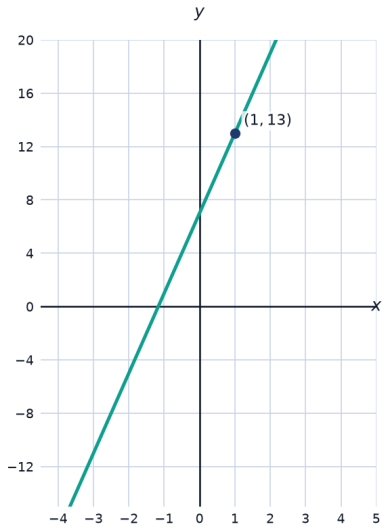
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 6$ in multiplying polynomials, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 7$ in special products, evaluate $f(2)$.



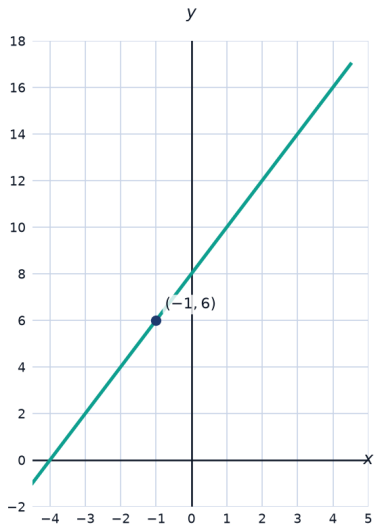
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for special products consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 8$ in factoring out the greatest common factor (gcf), evaluate $f(2)$.

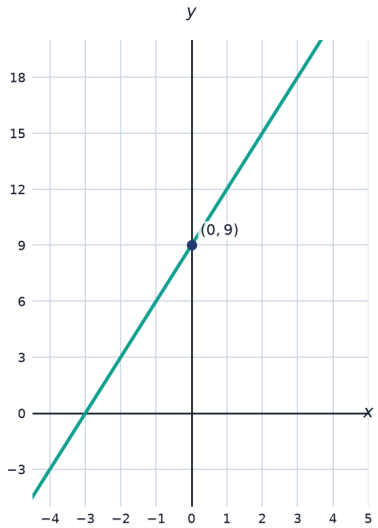


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for factoring out the greatest common factor (gcf) consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 3x + 9$ in factoring trinomials, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for factoring trinomials consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

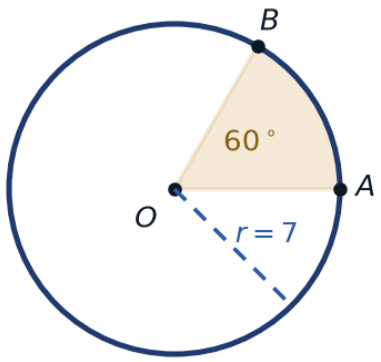
Solve $\sqrt{3x+1} = x - 1$, checking for extraneous solutions.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Radicals & Rational Exponents · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the circumference of a circle with $r = 7$. Take $\pi = \frac{22}{7}$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the equation $5e^{0.04t} = 45$ for t in terms of natural logarithms, and find its value rounded to two decimal places.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · AdvancedA school administrator randomly surveys 120 high school seniors at Oakwood High School. In the sample, 65% report planning to attend a four-year college, with a margin of error of $\pm 4\%$. Which of the following conclusions is most statistically sound? (A) Exactly 65% of all Oakwood High seniors will attend college. (B) It is plausible that between 61% and 69% of all Oakwood High seniors plan to attend a four-year college. (C) Between 61% and 69% of all high school seniors in the state plan to attend college. (D) To reduce the margin of error by half, the administrator should survey 240 seniors.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · AdvancedUsing the student prep table (Studied: 54 passed, 6 failed; Did not study: 18 passed, 22 failed; Total passed: 72 out of 100), are the events 'passed' and 'studied' independent?

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A cuboid has a length of USD $8cm, awidthofUSD5$ cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

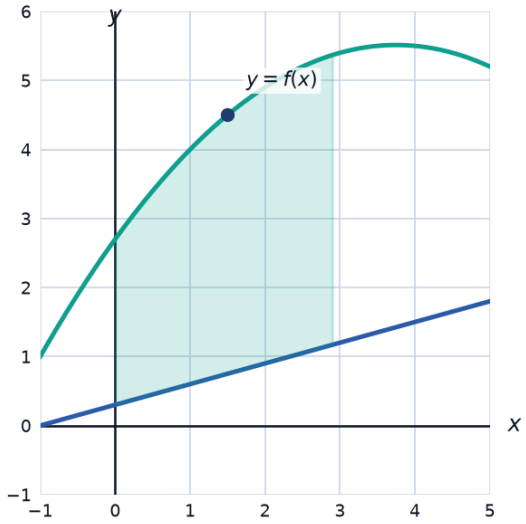
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Exponents & Polynomials

A line has slope 3 and passes through $(0, 2)$. Find y when $x = 2$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Functions & Transformations

Find the area of a triangle with base 10 and height 6.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Evaluate the complex expression $\frac{(3-2i)^2}{1+i}$ and write the result in standard form $a + bi$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Systems & Inequalities

In triangle XYZ, $XY = 7$ cm, $YZ = 10$ cm, and angle $XYZ = 65^\circ$. Calculate (a) the length of XZ to 3 significant figures and (b) the area of triangle XYZ.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

Solve the quadratic equation $2x^2 + 5x - 4 = 0$. Give your solutions correct to 2 decimal places.

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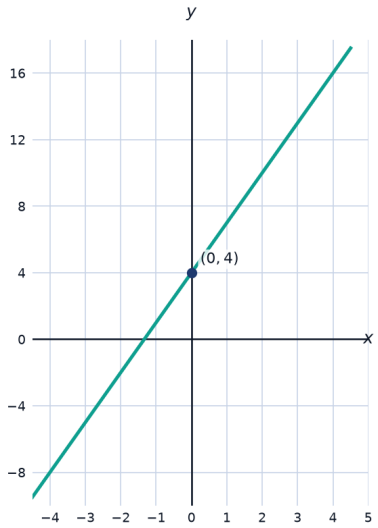
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Factoring Polynomials

Level 3 · AdvancedA quadratic function $f(x) = ax^2 + bx + c$ passes through the points $(-2, 11)$, $(0, 3)$, and $(2, 7)$. What is the value of $a + b + c$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in adding and subtracting polynomials, evaluate $f(2)$.



a. 10

b. 13

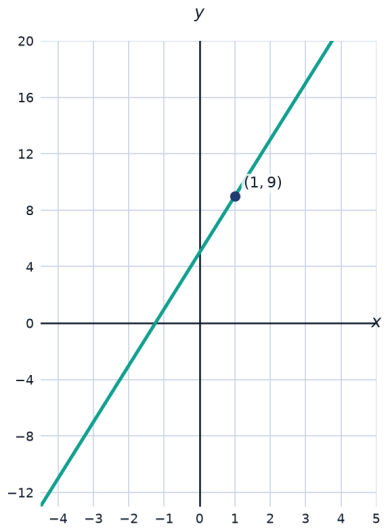
c. 6

d. 14

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in multiplying a polynomial by a monomial, evaluate $f(2)$.



a. 13

b. 17

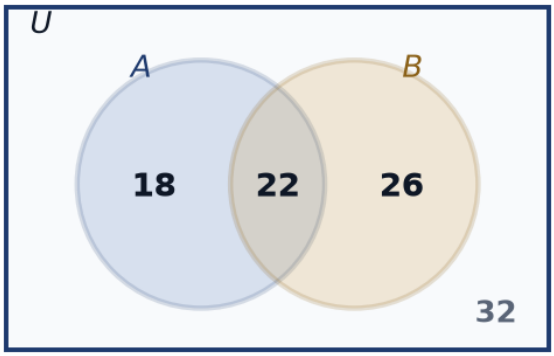
c. 8

d. 17

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for adding and subtracting polynomials consists of the four values $\{18, 22, 26, 32\}$.

What is the arithmetic mean?



a. 22.5

b. 23.5

c. 24.5

d. 26.5

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for multiplying a polynomial by a monomial consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For a constant k, the equation $\sqrt{2x + k} = x - 1$ has $x = 5$ as a valid solution. What is the extraneous solution produced when solving this equation by squaring both sides?

a. -5

b. -1

c. 1

d. 6

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the equation $\sqrt[3]{25 - x^2} + \sqrt[3]{x^2 + 3} = 4$ What is the product of all real solutions to this equation?

a. -24

b. 24

c. -48

d. 0

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $g(x) = \sqrt{x^2 - 6x + 9} + \sqrt{x^2 + 8x + 16}$. Find the minimum possible value of $g(x)$.

a. 2

b. 3

c. 4

d. 7

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which expression is equivalent to $(16x^8y^4)^{\frac{3}{4}}$ for all positive values of x and y?

a. $2x^2y$

b. $8x^6y^3$

c. $12x^6y^3$

d. $16x^6y^3$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the equation $\sqrt{2x + k} = x - 1$, where k is a constant. If $x = 5$ is a valid solution to the equation, what is the extraneous solution that arises when solving the equation algebraically by squaring both sides?

a. -1

b. 1

c. -5

d. 6

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = \sqrt{x^2 - 6x + 9}$ and $g(x) = x - 3$. For which values of x is $f(x) = g(x)$?

a. All real numbers

b. All real numbers $x \geq 3$

c. All real numbers $x \leq 3$

d. No real numbers

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the function $f(x) = a\sqrt{x - h} + k$ is shown below. The function passes through the points $(5, 10)$ and $(13, 16)$. Determine the value of a and the coordinates of the vertex (h, k).

a. $a = 3, (h, k) = (1, 1)$

b. $a = 2, (h, k) = (1, 4)$

c. $a = 3, (h, k) = (1, 4)$

d. $a = 2, (h, k) = (1, 1)$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the value of the expression $\sqrt{36} + \sqrt{49}$?

a. 10

b. 11

c. 13

d. 15

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Check: is $x = -1$ a solution of $\sqrt{x} = -1$?

a. Maybe

b. Only in $\mathbb{C}$

c. No (square root is non-negative)

d. Yes

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Write $x^{3/2}$ as a radical.

a. $x^{1/2} \cdot x$

b. $x\sqrt{x^2}$

c. $\sqrt{x^3}$ or $(\sqrt{x})^3$

d. $\sqrt[3]{x^2}$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find $f(x) = \sqrt{x} + \sqrt{1 - x}$. What is the domain?

a. $x \geq 0$

b. All reals

c. $0 < x < 1$

d. $0 \leq x \leq 1$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Write $\sqrt[3]{a^2 b}$ using rational exponents.

a. $a^{2/3} b^{1/3}$

b. $a^{1/2} b^{1/3}$

c. $a^{3/2} b^{1/3}$

d. $a^2 b^3$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following equals $8^{-2/3}$?

a. $\frac{1}{8}$

b. 4

c. $\frac{1}{4}$

d. -4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Systems & Inequalities

Find the real number(s) equal to $(-32)^{3/5}$.

a. 8

b. Undefined

c. -8

d. -32^3

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

Write $\sqrt[4]{x^3}$ using rational exponent.

a. x^{3+4}

b. $4x^3$

c. $x^{4/3}$

d. $x^{3/4}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Factoring Polynomials

Simplify $\sqrt{4}$ — note: solve $\sqrt{36}$.

a. 9

b. 6

c. 3

d. 12

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Linear Equations & Lines

Determine the number of real solutions of $\sqrt{x} = x^2 - 6x + 9$.

a. 0

b. 1

c. 4

d. 2

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Rational Expressions

Solve: $\sqrt{2x+3} = x$. Check for extraneous roots.

a. $x = 3$ only

b. $x = -1$ only

c. $x = 3$ and $x = -1$

d. No solution

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Statistics & Data Analysis

Prove $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$ using rational exponents.

a. True only for $n = 2$

b. $(ab)^{1/n} = a^{1/n}b^{1/n}$ — follows from power rules for real numbers

c. Not always true

d. Requires $a, b > 0$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratic Functions

Simplify: $\frac{\sqrt{x+h}-\sqrt{x}}{h}$ (rationalize numerator).

a. $\frac{1}{2\sqrt{x}}$

b. $\frac{1}{\sqrt{x+h}+\sqrt{x}}$

c. $\frac{h}{\sqrt{x+h}+\sqrt{x}}$

d. $\frac{\sqrt{x+h}-\sqrt{x}}{h}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Exponents & Polynomials

Express $\sqrt[6]{x^4}$ in simplest radical form.

a. $\sqrt[3]{x^2}$

b. $\sqrt[6]{x^4}$

c. $x^{2/3}$

d. Both A and C

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